Killiivalavan K A

killiivalavan.inbox@gmail.com | Killiivalavan | Portfolio | Killiivalavan

SUMMARY

Full-Stack developer experienced in building backend systems and AI-integrated applications including RAG pipelines, agentic automation, and async architecture. Comfortable across Python, Go, and TypeScript, with hands-on experience in self-hosted infrastructure and shipping production tools.

WORK EXPERIENCE

Yodigispace

Junior Programmer — Aug 2025 - Present

- Built a document Q&A RAG pipeline for the DMS module which includes page-aware chunking with a PDF extraction fallback chain (PyPDF, pdfplumber, Tesseract OCR), MiniLM embeddings in per-document ChromaDB collections, and cross-encoder reranking.
- Designed and built an entity-graph retrieval path (LLM batch entity extraction with spaCy fallback, NetworkX) fused with vector results via reciprocal rank fusion and built a 65-question eval set covering exact, paraphrase, multi-hop, and abstention cases.
- Designed a template-driven multi-step approval workflow with per-step assignees, transactional state transitions, version history, and notifications.
- Implemented per-tenant LLM configuration with Fernet-encrypted API keys, so each client company runs on its own provider key.
- Built HR (leave ledger, attendance, ECDSA-paired kiosk check-in) and Production (BOM release, job cards, QC, idempotent stock receipts) modules with transactional inventory integration.

TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, SQL, Golang
- Frameworks: React, Next.js, Tailwind CSS, FastAPI, Flask, Django, Node.js, Express.js
- Tools: Linux, Git, GitHub, Docker, Tailscale, Cloudflare Workers, pytest
- Data: PostgreSQL, SQLite, MongoDB, MySQL

PROJECTS

D.A.I.S.Y (2024 - Present)

- Built a fully asynchronous agentic voice assistant pipeline (OpenWakeWord, Silero VAD, Faster-Whisper, Kokoro) targeting sub 2 second end-to-end latency.
- Architected a client-server split over Tailscale, self-hosted on a home Ubuntu server, integrating 13+ tools enabling autonomous background coding sessions (via OpenCode), shell command execution, proactive task/reminder management, file I/O, and web search with text extraction
- Designed a role-based LLM router with per-role provider chains and automatic fallback across Groq, Cerebras, and Gemini.

DumpIt (2026 - Present)

- Built and shipped an Android journaling app (React Native, Expo, TypeScript) with Google sign-in via Supabase Auth (PKCE), distributed as an APK with a published SHA-256 checksum
- Designed offline-first sync - local-first storage with last-write-wins reconciliation against Postgres, row-level security on every query, and logout safeguards for unsynced data
- Built local reminder notifications, streak and statistics views, and a controversial neo-brutalistic UI and theme system

Omafinder (2026 - Present)

- Built and published an open-source (MIT) keyboard-first file manager/navigator plugin for Omarchy (Arch linux based OS), listed in the Omarchy plugin marketplace
- Implemented a custom fuzzy matcher with frecency ranking for file search, plus drive mounting via udisks2, previews, and multi-select operations

Also built: chesshell (Go terminal chess client with Stockfish and multiplayer) and **tabstone** (tab archiver live on the Chrome Web Store and Firefox Add-ons). More on my GitHub and portfolio.

EDUCATION

SRM University - B.tech CSE with specialization in AI and Data Science — 2021 - 2025

CERTIFICATIONS

Artificial Intelligence Analyst - IBM — June 2024

Introduction to Python - IBM — Nov 2023

Cloud Application Developer - IBM — June 2023